

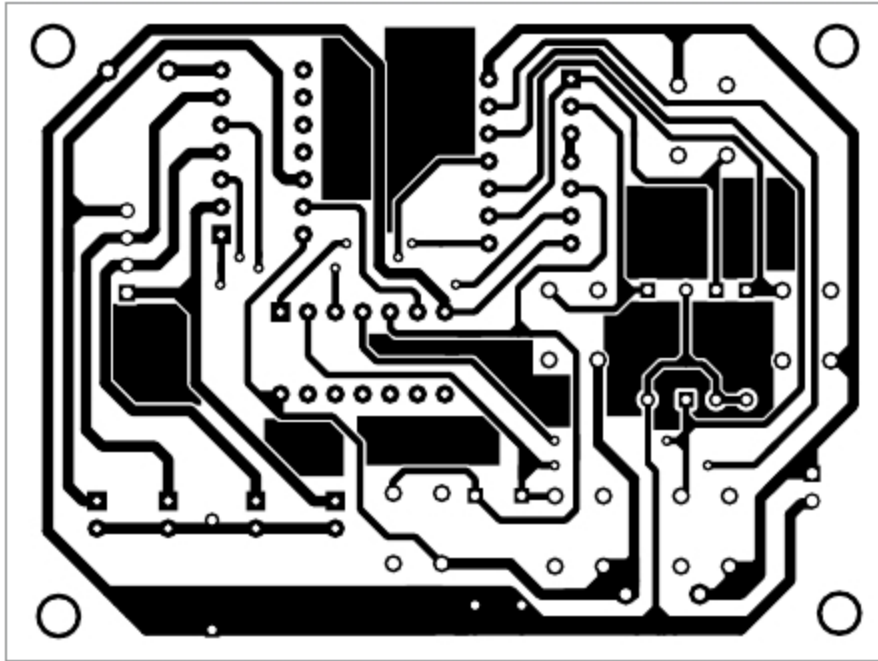


Fig. 3: Actual-size PCB layout of joystick for robot

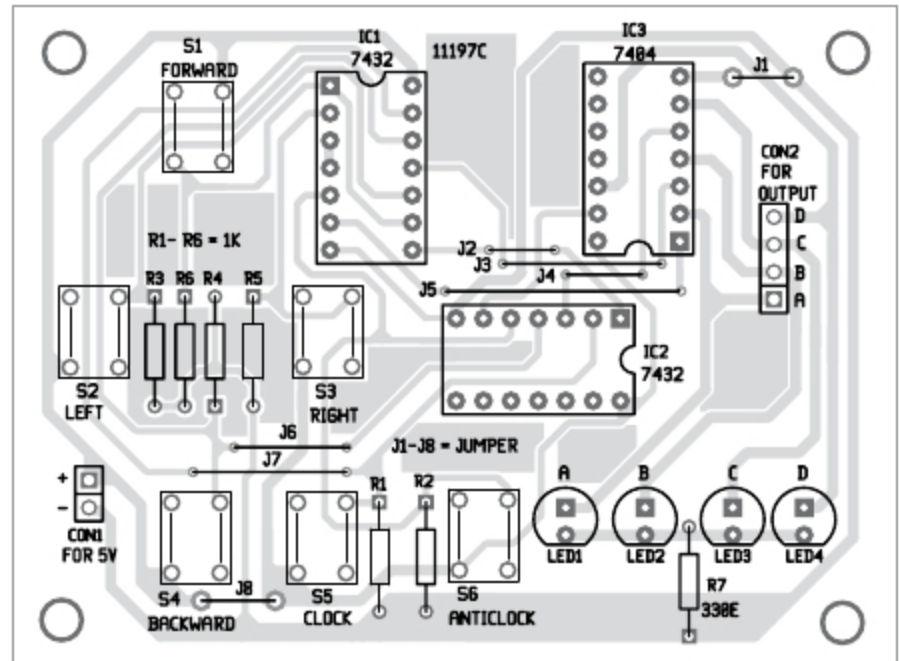


Fig. 4: Components layout for the PCB

TRUTH TABLE FOR SWITCHES AND LED STATUS

Pressed switch	Motion	A	B	C	D	LED1	LED2	LED3	LED4
S1	Forward	0	1	0	1	Off	On	Off	On
S2	Left	1	1	0	1	On	On	Off	On
S3	Right	0	1	1	1	Off	On	On	On
S4	Backward	1	0	1	0	On	Off	On	Off
S5	Clockwise	1	0	0	1	On	Off	Off	On
S6	Counter-clockwise	0	1	1	0	Off	On	On	Off

PARTS LIST

Semiconductors:

- IC1, IC2 - 7432 quad OR gate
- IC3 - 7404 hex inverter
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R6 - 1-kilo-ohm
- R7 - 330-ohm

Miscellaneous:

- CON1 - 2-pin connector
- CON2 - 4-pin connector
- S1-S6 - Tactile switches
- 5V regulated power supply

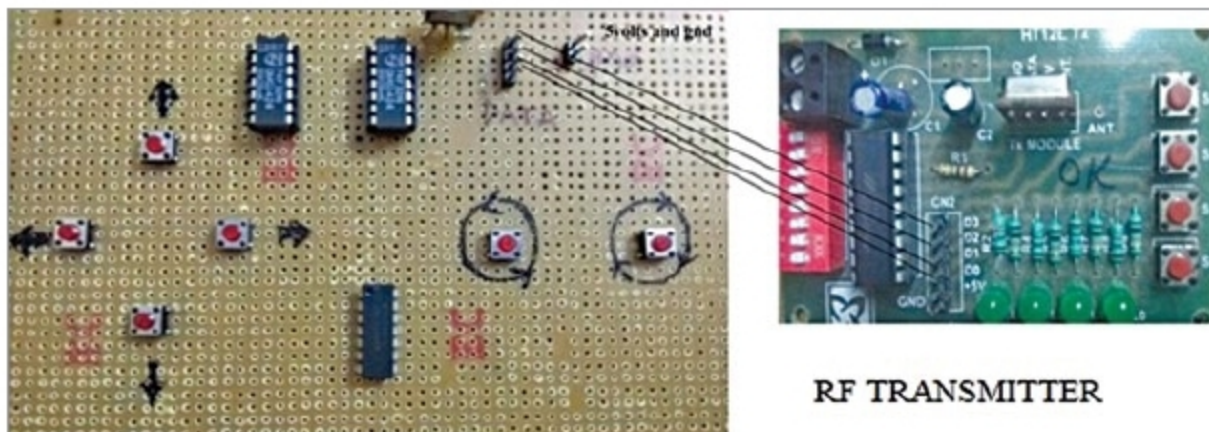


Fig. 5: Connections with RF transmitter

Connect all LEDs (LED1 through LED4) to the front panel for indicating output. LEDs will glow according to the code generated, indicating high (logic 1) or low (logic 0) status. That is, status of each input switch can be checked through the glowing of the

respective LED and the truth table.

You can feed the signals from CON2 either to input of the motor controller circuit to drive the robot or to input of the joystick to control the cursor in a video game application.

A readymade RF transmitter to

control the robot is easily available online or in the local market. The circuit can be used with the RF transmitter circuit to control the robot as shown in Fig. 5. Connect 4-pin header (A, B, C, D) of the joystick with 4-pin input (D0, D1, D2, D3) of the RF transmitter. You will find a 6-pin header in the RF transmitter marked as D0, D1, D2, D3, +5V and GND. If you are using a separate supply for joystick, do not forget to make ground common. **EFY**



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